INCH-POUND

The documentation and process conversion measures necessary to comply with this revision shall be completed by 21 August 2021.

MIL-PRF-19500/767C
W/AMENDMENT 1
21 April 2021
SUPERSEDING
MIL-PRF-19500/767C
8 April 2016

PERFORMANCE SPECIFICATION SHEET

TRANSISTOR, NPN, SILICON, LOW-POWER, TYPE 2N5551,
ENCAPSULATED (SURFACE MOUNT PACKAGES) AND UNENCAPSULATED,
RADIATION HARDNESS ASSURANCE,
QUALITY LEVELS JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for NPN silicon, low-power transistors. Four levels of product assurance (JAN, JANTX, JANTXV, and JANS) are provided for each device type as specified in MIL-PRF-19500 and two levels of product assurance (JANHC and JANKC) are provided for each unencapsulated type die. Provision for radiation hardness assurance RHA level designators "M", "D", "P", "L", "R", "F", "G", and "H" are appended to the device prefix to identify devices which have passed RHA requirements.

1.2 Package outlines. The device package outline is an unleaded four terminal surface mount (UB) in accordance with figure 1.

1.3 Maximum ratings. Unless otherwise specified, T_A = +25 °C.

Type	P_{T1} (1) (2) T_C = +25°C	P_{T2} (1) T_A = +25°C	P_{T3} (1) $T_{SP(IS)}$ = +25°C	$R_{\theta JA}$	$R_{\theta JSP(IS)}$	V_{CBO}	V_{CEO}	V_{EBO}	I_C	T_J and T_{STG}
	W	W	W	°C/W	°C/W	V dc	V dc	V dc	mA dc	°C
2N5551	1.2	0.36	1	280	85	180	160	6	500	-65 to +200

(1) For thermal impedance curves for 2N5551, see figures 2 and 3 for T_A > +25 °C.
(2) For variants with tin-lead plating or hot solder dip lead finish all testing, and any handling, performed at T_A > +125 °C shall be carried out in a 100 percent inert atmosphere.

Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A

FSC 5961

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1.4 Primary electrical characteristics.

Limits	h_{FE1} (1) V_{ce} = 5 V dc I_C = 1 mA dc	h_{FE2} (1) V_{ce} = 5 V dc I_C = 10 mA dc	h_{FE3} (1) V_{ce} = 5 V dc I_C = 50 mA dc	$\lvert h_{fe} \rvert$ V_{ce} = 10 V dc I_C = 10 mA dc f = 20 MHz	$V_{CE(SAT)1}$ (1) I_C = 10 mA dc I_B = 1.0 mA dc	$V_{BE(SAT)}$ (1) I_C = 10 mA dc I_B = 1.0 mA dc
					mV dc	mV dc
Min	80	80	30	2.5		
Max		250			150	1

(1) Pulsed (see 4.5.1).

1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.4 for PIN construction example and 6.5 for a list of available PINs.

1.5.1 JAN certification mark and quality level.

1.5.1.1 Quality level designators for encapsulated devices. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: "JAN", "JANTX", "JANTXV", and "JANS".

1.5.1.2 Quality level designators for un-encapsulated devices (die). The quality level designators for un-encapsulated devices (die) that are applicable for this specification sheet from the lowest to the highest level are as follows: "JANHC" and "JANKC".

1.5.2 Radiation hardness assurance (RHA) designator. The RHA levels that are applicable for this specification sheet from lowest to highest are as follows: "M", "D", "P", "L", "R", "F", "G", and "H".

1.5.3 Device type. The designation system for the device types covered by this specification sheet are as follows.

1.5.3.1 First number and first letter symbols. The transistors of this specification sheet use the first number and letter symbols "2N".

1.5.3.2 Second number symbols. The second number symbols for the transistors covered by this specification sheet are as follows: "5551".

1.5.4 Suffix symbols. The following suffix letters are incorporated in the PIN for this specification sheet.

UB	Indicates an unleaded four terminal surface mount package (see figure 1)

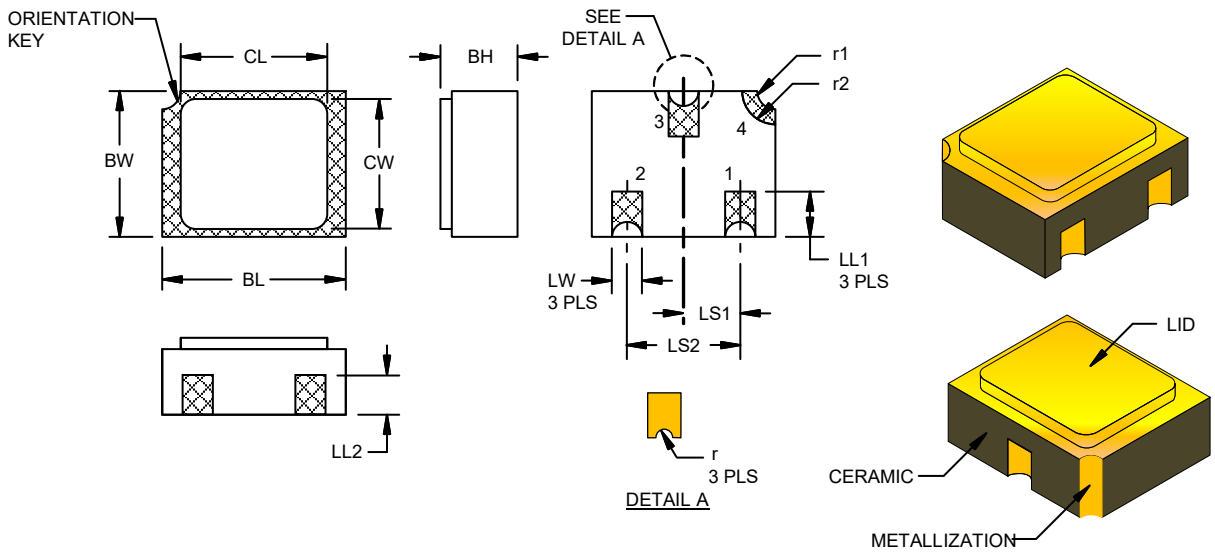
1.5.5 Lead finish. The lead finishes applicable to this specification sheet are listed on QML-19500.

1.6 Radiation features: The maximum total ionizing dose (TID) available for RHA devices is in accordance with condition A (dose rate = 50 to 300 rad(Si)/s) of method 1019 of MIL-STD-750:

For device type JANSR2N5551UB: 100 krads(Si) 1/

1/ Manufacturer's supplying device type 2N75551UB has performed characterization testing in accordance with test condition A of method 1019 of MIL-STD-750 (dose rate = 50 to 300 rad(Si)/s). The radiation end point limits are guaranteed only for a maximum TID level of 100 krads(Si)/s.

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Symbol	Dimensions				Symbol	Dimensions			
	Inches		Millimeters			Inches		Millimeters	
	Min	Max	Min	Max		Min	Max	Min	Max
BH	.046	.056	1.17	1.42	LS1	.035	.039	0.89	0.99
BL	.115	.128	2.92	3.25	LS2	.071	.079	1.80	2.01
BW	.085	.108	2.16	2.74	LW	.016	.024	0.41	0.61
CL		.128		3.25	r	.008 TYP		0.20 TYP	
CW		.108		2.74	r1	.012 TYP		0.30 TYP	
LL1	.022	.038	0.56	0.96	r2	.022 TYP		0.56 TYP	
LL2	.017	.035	0.43	0.89					

NOTES:
1. Dimensions are in inches. 2. Millimeters are given for general information only.
3. Hatched areas on package denote metallized areas.
4. Pad 1 = Base, Pad 2 = Emitter, Pad 3 = Collector, Pad 4 = Shielding connected to the lid.
5. Dimensions are pre-solder dip.
6. In accordance with ASME Y14.5M, diameters are equivalent to Øx symbology.

FIGURE 1. Physical dimensions, surface mount (2N5551UB).

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 – Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 – Test Methods for Semiconductor Devices.

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows.

$T_{SP(IS)}$ Temperature solder pad (infinite sink).

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figure 1.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Marking. Devices shall be marked in accordance with MIL-PRF-19500, except for the UB suffix package. Marking on the UB package shall consist of an abbreviated part number, the date code, and the manufacturer's symbol or logo. The prefixes JAN, JANTX, JANTXV, and JANS can be abbreviated as J, JX, JV, and JS respectively. The radiation hardened designator M, D, P, L, R, F, G, or H shall immediately precede (or replace) the device "2N" identifier (depending upon degree of abbreviation required).

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

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3.7 Electrical test requirements. The electrical test requirements shall be as specified in table I.

3.8 RHA. RHA requirements and test levels shall be as defined in MIL-PRF-19500 and herein. RHA characterization data shall be made available upon request of the qualifying or acquiring activity.

3.8.1 TID. When TID testing is performed as in accordance with 1.6, the device shall be capable of meeting the radiation end-point electrical limits listed in table II.

3.8.2 Displacement damage characterization (only when specified, see 6.2.e). When displacement damage characterization is performed, it shall demonstrate exposure versus data and shall not be used to indicate pass or fail criteria.

3.9 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and table I and table II).

d. Element evaluation (see 4.6).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table III tests, the tests specified in table III herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

4.2.2 RHA qualification. RHA qualification is required only for those devices with the RHA designator as specified herein.

4.2.2.1 TID. TID testing shall be in accordance with condition A of method 1019 of MIL-STD-750 using the dose rate specified in 3.8.1.

4.2.2.3 Displacement damage characterization. The displacement damage exposure shall be conducted in accordance with method 1017 of MIL-STD-750. The following details shall apply:

a. Samples may be taken from any wafer of the qualification lot.

b. As a minimum, testing shall be at 2^{12} n/cm^2 plus two additional neutron fluence levels chosen by the manufacturer.

c. If the device degrades less than 5 percent of the specification at the highest neutron fluence level, a single data point may be sufficient.

Alternate package options may be substituted for displacement damage characterization. The displacement damage characterization data shall be made available from the manufacturer.

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4.3 Screening (quality levels JANS, JANTX, and JANTXV). Screening shall be in accordance with table E-IV of MIL-PRF-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen	Measurement	
	JANS	JANTX and JANTXV levels
(1) 3c	Thermal impedance (see 4.3.2).	Thermal impedance (see 4.3.2).
7	Optional	Optional
9	I_{CBO1} h_{FE2}	Not applicable
11	I_{CBO1} h_{FE2} ΔI_{CBO1} = 100 percent of initial value or 5 nA dc, whichever is greater; Δh_{FE2} = 15 percent of initial value.	I_{CBO1} h_{FE2}
12	See 4.3.1.	See 4.3.1.
13	ΔI_{CBO1} = 100 percent of initial value or 5 nA dc, whichever is greater; Δh_{FE2} = 15 percent of initial value, subgroup 2 and 3 of table I herein.	ΔI_{CBO1} = 100 percent of initial value or 5 nA dc, whichever is greater; Δh_{FE2} = 15 percent of initial value, subgroup 2 of table I herein.
14	Required	Required

(1) Shall be performed anytime after temperature cycling, screen 3a; Quality levels JANTX and JANTXV do not need to be repeated in screening requirements.

4.3.1 Power burn-in conditions. Power burn-in conditions shall be as follows: V_{CB} = 10 to 30 V dc. Power shall be applied to achieve T_J = +135 °C minimum using a minimum P_D = 75 percent of P_T maximum rated as defined in 1.3. With approval of the qualifying activity and preparing activity, alternate burn-in criteria (hours, bias conditions, T_J, and mounting conditions) may be used for JANTX and JANTXV quality levels. A justification demonstrating equivalence is required. In addition, the manufacturing site's burn-in data and performance history will be essential criteria for burn-in modification approval.

4.3.2 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3131 of MIL-STD-750 using the guidelines in that method for determining I_M, I_H, t_H, t_{SW} (V_C and V_H where appropriate). See group E, subgroup 4, table III herein.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500 and table I herein.

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4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with 4.4.2.1 for quality level JANS and 4.4.2.2 for quality levels JAN, JANTX, and JANTXV.

4.4.2.1 Quality level JANS, table E-VIA of MIL-PRF-19500. Group B inspection for quality level JANS shall be conducted in accordance with the conditions specified for subgroup testing in table E-VIA of MIL-PRF-19500 and herein. Delta measurements shall be in accordance with 4.7 herein.

Subgroup	Method	Condition
B4	1037	V_{CB} = 10 V dc, adjust device current, or power, to achieve a minimum ΔT_J of +100 °C.
B5	1027	V_{CB} = 10 V dc; $P_D \geq$ 100 percent of maximum rated P_T (see 1.3). (NOTE: If a failure occurs, resubmission shall be at the test conditions of the original sample.) Option 1: 96 hours minimum sample size in accordance with MIL-PRF-19500, table E-VIA, adjust T_A or P_D to achieve T_J = +275 °C minimum. Option 2: 216 hours minimum, sample size = 45, c = 0; adjust T_A or P_D to achieve a T_J = +225°C minimum.

4.4.2.2 Quality levels JAN, JANTX, and JANTXV, table E-VIC of MIL-PRF-19500. Group B inspection for quality levels JAN, JANTX, and JANTXV shall be conducted in accordance with the conditions specified in table E-VIC (small die flow) of MIL-PRF-19500 and herein. Delta measurements shall be taken after each step and shall be in accordance with 4.7 herein.

Step	Method	Condition
1	1026	Steady-state life: 1,000 hours minimum, V_{CB} = 10 V dc, power shall be applied to achieve T_J = +150 °C minimum using a minimum of P_D = 75 percent of maximum rated P_T as defined in 1.3. n = 45 devices, c = 0. The sample size may be increased and the test time decreased as long as the devices are stressed for a total of 45,000 device hours minimum, and the actual time of test is at least 340 hours.
2	1048	Blocking life, T_A = 150 °C, V_{CB} = 80 percent of rated voltage, 48 hours minimum. n = 45 devices, c = 0.
3	1032	High-temperature life (non-operating), t = 340 hours T_A = +200 °C. n = 22, c = 0.

4.4.2.2.1 Group B sample selection. Samples selected for small die flow group B inspection shall be in accordance with the following requirements:

a. For JAN, JANTX, and JANTXV samples shall be selected randomly from a minimum of three wafers (or from each wafer in the lot) from each wafer lot.

b. Shall be chosen from an inspection lot that has been submitted to and passed table I, subgroup 2, conformance inspection.

c. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for life test (step 1) may be tested prior to the application of final lead finish.

d. Separate samples may be used for each step.

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4.4.2.2.2 Catastrophic failures. All catastrophic failures during CI shall be analyzed to the extent possible to identify root cause and corrective action. Whenever a failure is identified as wafer lot or wafer processing related, the entire wafer lot and related devices assembled from the wafer lot shall be rejected unless an appropriate determined corrective action to eliminate the failure mode has been implemented and the devices from the wafer lot are screened to eliminate the failure mode.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the tests and conditions specified for subgroup testing in table E-VII of MIL-PRF-19500, and in 4.4.3.1 for quality level JANS and 4.4.3.2 (for quality levels JAN, JANTX, and JANTXV herein for group C testing. Delta measurements are in accordance with 4.6 herein.

4.4.3.1 Quality level JANS, table E-VII of MIL-PRF-19500.

Subgroup	Method	Condition
C5	3131	$R_{\theta JA}$ and $R_{\theta JC}$ only, as applicable (see 1.3 and 4.3.2) and in accordance with thermal impedance curves.
C6	1026	1,000 hours at V_{CB} = 10 V dc; power shall be applied to achieve T_J = +150 °C minimum and a minimum of P_D = 75 percent of maximum rated P_T as defined in 1.3 n = 45, c = 0. The sample size may be increased and the test time decreased as long as the devices are stressed for a total of 45,000 device hours minimum, and the actual time of test is at least 340 hours.

4.4.3.2 Quality levels JAN, JANTX, and JANTXV, table E-VII of MIL-PRF-19500.

Subgroup	Method	Condition
C5	3131	$R_{\theta JA}$ and $R_{\theta JC}$ only, as applicable (see 1.3 and 4.3.2) and in accordance with thermal impedance curves.
C6	1026	Not applicable.

4.4.3.3 Group C sample selection. Samples for subgroups in group C inspection shall be chosen at random from any inspection lot containing the intended package type and lead finish procured to the same specification which is submitted to and passes group A tests for conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for C6 life test may be tested prior to the application of final lead finish. Testing of a group using a single device type enclosed in the intended package type shall be considered as complying with the requirements for that subgroup.

4.4.4 Group D inspection. Conformance inspection for radiation hardness assured JANS and JANTXV types shall include the group D tests specified in table II herein. These tests shall be performed as required in accordance with MIL-PRF-19500 and method 1019 of MIL-STD-750, for total ionizing dose or method 1017 of MIL-STD-750 for neutron fluence as applicable (see 6.2.e herein), except group D, subgroup 2 may be performed separate from other subgroups. Alternate package options may also be substituted for the testing provided there is no adverse effect to the fluence profile.

4.4.5 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-IX of MIL-PRF-19500 and as specified in table III herein. Delta requirements are in accordance with 4.6 herein.

4.5 Method of inspection. Methods of inspection shall be as specified in appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurements shall be as specified in section 4 of MIL-STD-750. Pulse measurement: Pulse width ≤ 300µs, duty cycle ≤ 2 percent.

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4.6 Element evaluation of unencapsulated die. The element evaluation of unencapsulated die shall be in accordance with appendix G of MIL-PRF-19500. For subgroup 4, the burn-in duration for the JANKC level shall follow the JANS requirements and the JANHC shall follow the JANTX requirements.

4.7 Delta measurements. Delta measurements shall be as specified below:

Step	Inspection	MIL-STD-750		Symbol	Limit	Unit
		Method	Conditions			
1.	Collector-base cutoff current	3036	Bias condition D, V_{CB} = 120 V dc	ΔI_{CBO1}	±100 percent of initial value or 5 nA dc, whichever is greater	
2.	Forward current transfer ratio	3076	V_{CE} = 5 V dc, I_C = 10 mA dc	ΔH_{FE2}	±15 percent of initial value	
3.	Collector-emitter saturation voltage	3071	I_C = +50 mA dc, I_B = 5 mA dc	$\Delta V_{CE(sat)2}$	±15 percent or 30 mV whichever is greater	

TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1 2/						
Visual and mechanical examination	2071					
Solderability 3/	2026	n = 15 leads, c = 0				
Resistance to solvents 3/ 4/	1022	n = 15 devices, c = 0				
Temp cycling 3/	1051	Test condition C, 25 cycles. n = 22 devices, c = 0				
Hermetic seal 3/ 5/	1071	n = 22 devices, c = 0				
Electrical measurements 3/		Table I, subgroup 2				
Bond strength 3/	2037	Precondition T_A = +250 °C at t = 24 hours or T_A = +300 °C at t = 2 hours, n = 11 wires, c = 0				
Decap internal visual (design verification) 3/	2075	n = 4 devices, c = 0				

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 2						
Thermal impedance 6/	3131	See 4.3.2	$Z_{\theta JX}$			°C/W
Collector to base cutoff current	3036	Bias condition D, V_{CB} = 120 V dc; pulsed (see 4.5.1)	I_{CBO1}		50	nA dc
Emitter to base cutoff current	3061	Bias condition D, V_{EB} = 4 V dc; pulsed (see 4.5.1)	I_{EBO1}		50	nA dc
Breakdown voltage, Collector to base	3001	Bias condition D; I_C = 100 µA dc; pulsed (see 4.5.1)	$V_{(BR)CBO}$	180		V dc
Breakdown voltage, collector to emitter	3011	Bias condition D; I_C = 1 mA dc; pulsed (see 4.5.1)	$V_{(BR)CEO}$	160		V dc
Breakdown voltage, emitter to base	3026	Bias condition D, I_E = 10 µA dc, pulsed (see 4.5.1)	$V_{(BR)EBO}$	6		V dc
Forward-current transfer ratio	3076	V_{CE} = 5 V dc, I_C = 1 mA dc, pulsed (see 4.5.1)	h_{FE1}	80		
Forward-current transfer ratio	3076	V_{CE} = 5 V dc, I_C = 10 mA dc, pulsed (see 4.5.1)	h_{FE2}	80	250	
Forward-current transfer ratio	3076	V_{CE} = 5 V dc, I_C = 10 mA dc, pulsed (see 4.5.1)	h_{FE2}	80	250	
Forward-current transfer ratio	3076	V_{CE} = 5 V dc, I_C = 50 mA dc, pulsed (see 4.5.1)	h_{FE3}	30		
Collector to emitter saturated voltage	3071	I_C = 10 mA dc, I_B = 1 mA dc, pulsed (see 4.5.1)	$V_{CE(sat)1}$		150	mV dc
Collector to emitter saturated voltage	3071	I_C = 50 mA dc, I_B = 5 mA dc, pulsed (see 4.5.1)	$V_{CE(sat)2}$		200	mV dc
Base to emitter saturated voltage	3066	Test condition A, I_C = 10 mA dc, I_B = 1 mA dc, pulsed (see 4.5.1)	$V_{BE(sat)1}$		1	V dc
Base to emitter saturated voltage	3066	Test condition A, I_C = 50 mA dc, I_B = 5 mA dc, pulsed (see 4.5.1)	$V_{BE(sat)2}$		1	V dc

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 3						
High temperature operation:		T_A = +150 °C				
Collector to base cutoff current	3036	Bias condition D, V_{CB} = 120 V dc; T_A = 150 °C (+0, -5)	I_{CBO2}		50	µA dc
Low temperature operation		T_A = -55 °C				
Forward-current transfer ratio	3076	V_{CE} = 5 V dc, I_C = 10 mA dc, pulsed (see 4.5.1)	h_{FE4}	20		
Subgroup 4						
Small signal short circuit forward current transfer ratio	3206	V_{CE} = 10 V dc, f = 20 MHz, I_C = 10 mA dc:	h_{fe}	2.5		
Open-circuit output capacitance	3236	V_{CB} = 10 V dc, I_E = 0, f = 1 MHz	C_{obo}		6	pF
Input capacitance (output open circuit output or short-circuit	3240	V_{EB} = 500 mV, I_C = 0 A f = 1 MHz	C_{ebo}		45	pF
Subgroups 5, 6, and 7						
Not applicable						

1/ For sampling plan see MIL-PRF-19500.
2/ For resubmission of failed test in subgroup 1 of table I, double the sample size of the failed test or sequence of tests. A failure in table I, subgroup 1 shall not require retest of the entire subgroup. Only the failed test shall be rerun upon submission.
3/ Not required for JANS.
4/ Not required for laser marked devices.
5/ This hermetic seal test is an end-point to temp-cycling in addition to electrical measurements.
6/ This test required for the following end-point measurements only:
Group B, subgroups 3, 4 and 5 (JANS).
Group B, step 1(JAN, JANTX, JANTXV).
Group C, subgroups 2 and 6.
Group E, subgroup 1.

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TABLE II. Group D inspection and end-point limits.

Inspection 1/ 2/ 3/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1 4/						
Neutron irradiation	1017	Neutron exposure V_{CES} = 0 V				
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 120 V dc; pulsed (see 4.5.1)	I_{CBO1}		100	nA dc
Emitter to base cutoff current	3061	Bias condition D; V_{EB} = 4.0 V dc; pulsed (see 4.5.1)	I_{EBO1}		100	nA dc
Breakdown voltage, collector to emitter	3011	Bias condition D; I_C = 1 mA dc; pulsed (see 4.5.1)	$V_{(BR)CEO}$	160		V dc
Breakdown voltage, collector to base	3001	Bias condition D I_C = 100 µA dc	$V_{(BR)CBO}$	180		V dc
Breakdown voltage, emitter to base	3026	Bias condition D, I_E = 10 µA dc	$V_{(BR)EBO}$	6		V dc
Forward-current transfer ratio	3076	V_{CE} = 5 V dc, I_C = 1 mA dc, pulsed (see 4.5.1)	[h_{FE1}] 5/	[40]		
Forward-current transfer ratio	3076	V_{CE} = 5 V dc, I_C = 10 mA dc, pulsed (see 4.5.1)	[h_{FE2}] 5/	[40]	250	
Forward-current transfer ratio	3076	V_{CE} = 5 V dc, I_C = 50 mA dc, pulsed (see 4.5.1)	[h_{FE3}] 5/	[15]		
Collector to emitter saturated voltage	3071	I_C = 10 mA dc; I_B = 1 mA dc; pulsed (see 4.5.1)	$V_{CE(sat)1}$		172.5	mV dc
Collector to emitter saturated voltage	3071	I_C = 50 mA dc; I_B = 5 mA dc; pulsed (see 4.5.1)	$V_{CE(sat)2}$		230	mV dc
Base to emitter saturated voltage	3066	Test condition A; I_C = 10 mA dc; I_B = 1 mA dc, pulsed (see 4.5.1)	$V_{BE(sat)1}$		1.15	V dc
Base to emitter saturated voltage	3066	Test condition A; I_C = 50 mA dc; I_B = 5 mA dc, pulsed (see 4.5.1)	$V_{BE(sat)2}$		1.15	V dc
Subgroup 2						
Steady-state total dose irradiation	1019	Gamma exposure V_{CES} = 128 V				

See footnotes at end of table.

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TABLE II. Group D inspection and end-point limits - Continued.

Inspection 1/ 2/ 3/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 2 – Continued						
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 120 V dc; pulsed (see 4.5.1)	I_{CBO1}		100	nA dc
Emitter to base cutoff current	3061	Bias condition D; V_{EB} = 4.0 V dc; pulsed (see 4.5.1)	I_{EBO1}		100	nA dc
Breakdown voltage, collector to emitter	3011	Bias condition D; I_C = 1 mA dc; pulsed (see 4.5.1)	$V_{(BR)CEO}$	160		V dc
Breakdown voltage, collector to base	3001	Bias condition D I_C = 100 µA dc	$V_{(BR)CBO}$	180		V dc
Breakdown voltage, emitter to base	3026	Bias condition D, I_E = 10 µA dc	$V_{(BR)EBO}$	6		V dc
Forward-current transfer ratio	3076	V_{CE} = 5 V dc, I_C = 1 mA dc, pulsed (see 4.5.1)	[h_{FE1}] 5/	[40]		
Forward-current transfer ratio	3076	V_{CE} = 5 V dc, I_C = 10 mA dc, pulsed (see 4.5.1)	[h_{FE2}] 5/	[40]	250	
Forward-current transfer ratio	3076	V_{CE} = 5 V dc, I_C = 50 mA dc, pulsed (see 4.5.1)	[h_{FE3}] 5/	[15]		
Collector to emitter saturated voltage	3071	I_C = 10 mA dc; I_B = 1 mA dc; pulsed (see 4.5.1)	$V_{CE(sat)1}$		172.5	mV dc
Collector to emitter saturated voltage	3071	I_C = 50 mA dc; I_B = 5 mA dc; pulsed (see 4.5.1)	$V_{CE(sat)2}$		230	mV dc
Base to emitter saturated voltage	3066	Test condition A; I_C = 10 mA dc; I_B = 1 mA dc, pulsed (see 4.5.1)	$V_{BE(sat)1}$		1.15	V dc
Base to emitter saturated voltage	3066	Test condition A; I_C = 50 mA dc; I_B = 5 mA dc, pulsed (see 4.5.1)	$V_{BE(sat)2}$		1.15	V dc

1/ Tests to be performed on all devices receiving radiation exposure.
2/ For sampling plan, see MIL-PRF-19500.
3/ Electrical characteristics apply to the corresponding UB suffix version unless otherwise noted.
4/ See 6.2.e herein.
5/ See method 1019 of MIL-STD-750, for how to determine [h_{FE}] by first calculating the delta ($1/h_{FE}$) from the pre- and post-radiation h_{FE}. Notice the [h_{FE}] is not the same as h_{FE} and cannot be measured directly. The [h_{FE}] value can never exceed the pre-radiation minimum h_{FE} that it is based upon.

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TABLE III. Group E inspection (all quality levels) - for qualification or re-qualification only.

Inspection	MIL-STD-750		Sample plan
	Method	Conditions	
Subgroup 1			45 devices, c = 0
Temperature cycling (air to air)	1051	Test condition C, 500 cycles.	
Hermetic seal	1071	Fine and gross leak.	
Electrical measurements		See table I, subgroup 2 herein.	
Subgroup 2			45 devices, c = 0
Intermittent life	1037	V_{CB} = 10 V dc, 6,000 cycles. Adjust device current, or power, to achieve a minimum ΔT_J of +100 °C.	
Electrical measurements		See table I, subgroup 2 herein.	
Subgroup 4			
Thermal resistance	3131	$R_{\theta JSP(IS)}$ can be calculated but shall be measured once in the same package with a similar die size to confirm calculations (may apply to multiple specification sheets).	15 devices, c = 0
Thermal impedance curves		See table E-IX of MIL-PRF-19500, group E, subgroup 4.	Sample size N/A
Subgroup 5			
Not applicable			
Subgroup 6			
ESD	1020		
Subgroup 8			45 devices, c = 0
Reverse stability	1033	Condition B.	

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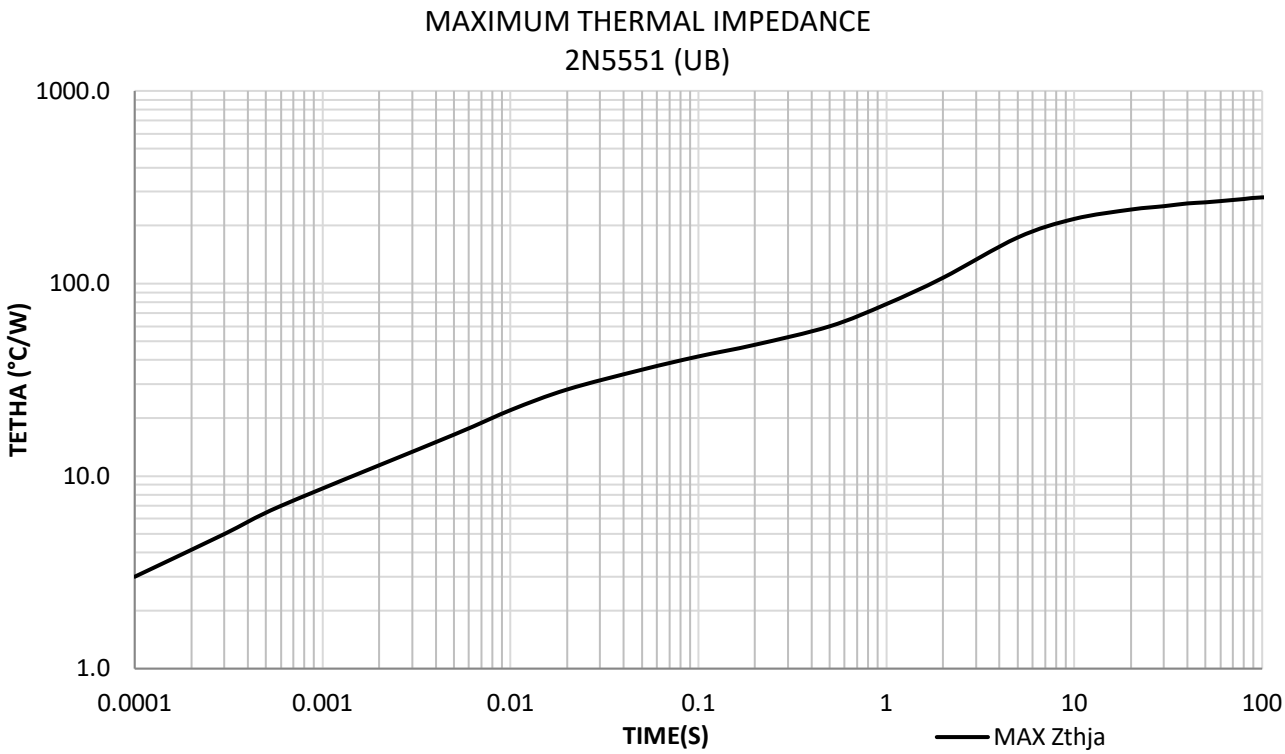


FIGURE 2. Thermal impedance graph ($R_{\theta JA}$) for 2N5551UB.

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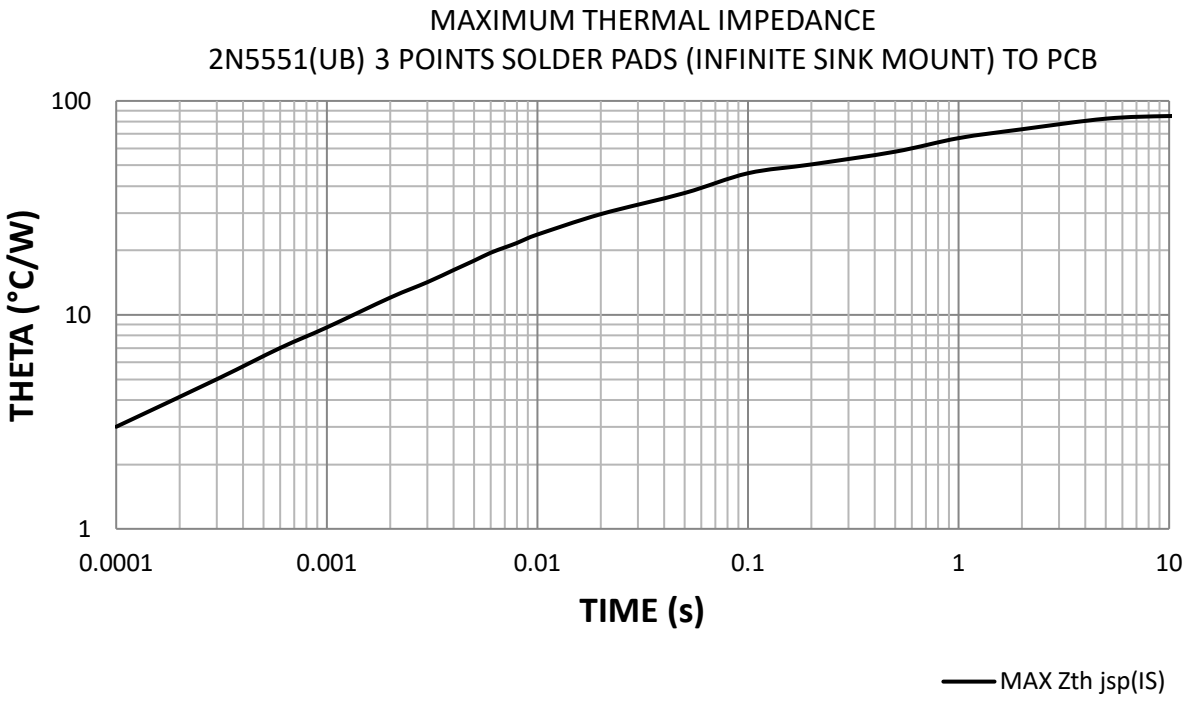


FIGURE 3. Thermal impedance graph ($R_{\theta JSP}$) for 2N5551UB.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

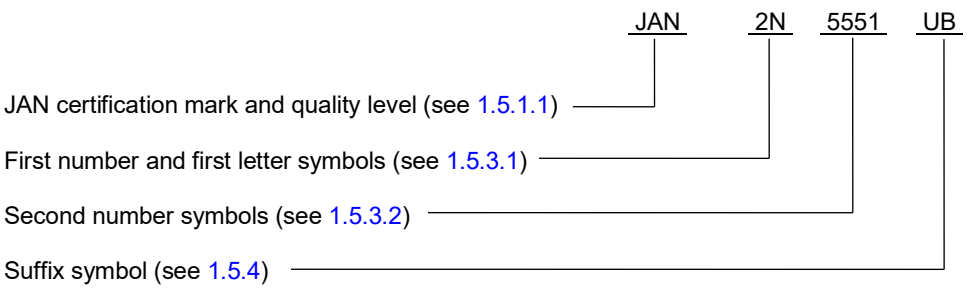
d. The complete PIN, see 1.5.

e. For acquisition of RHA designed devices, table II, subgroup 1 testing of group D is optional. If subgroup 1 testing is desired, it must be specified in the contract.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://qpldocs.dla.mil.

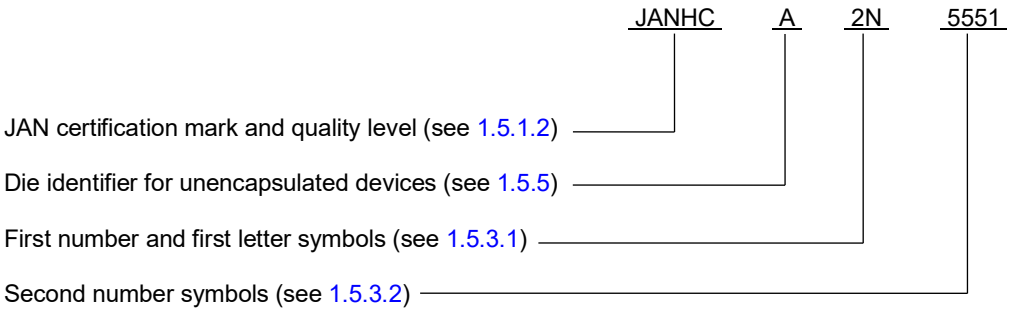
6.4 PIN construction example.

6.4.1 Encapsulated devices. The PINs for encapsulated devices are constructed using the following form.



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6.4.2 Un-encapsulated devices. The PINs for un-encapsulated devices are constructed using the following form.



6.5 List of PINs.

6.5.1 PINs for encapsulated devices. The following is a list of possible PINs for encapsulated devices available on this specification sheet.

PINs for type 2N5551			
JAN2N5551UB	JANTX2N5551UB	JANTXV2N5551UB	JANS#2N5551UB

(1) The number sign (#) represent one of eight RHA designators available (M, D, P, L, R, F, G, or H). The PIN is also available without a RHA designator.

6.5.2 PINs for un-encapsulated devices. The following is a list of possible PINs for un-encapsulated devices available on this specification sheet.

JANHCA2N5551	JANKCA2N5551

6.6 Suppliers of JANHC and JANKC die. The qualified JANHC and JANKC suppliers with the applicable letter version (example JANHCA2N5551) will be identified on the QML.

Die ordering information	
PIN	Manufacturer
	F8859
2N5551	JANHCA2N5551 JANKCA2N5551

6.7 Amendment notations. The margins of this specification are marked with asterisks to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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Custodians:
Army − CR
Navy − SH
Air Force − 85
NASA − NA
DLA − CC

Review activities:
Army − AR, MI, SM
Navy − AS, MC
Air Force − 19

Preparing activity:
DLA − CC

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